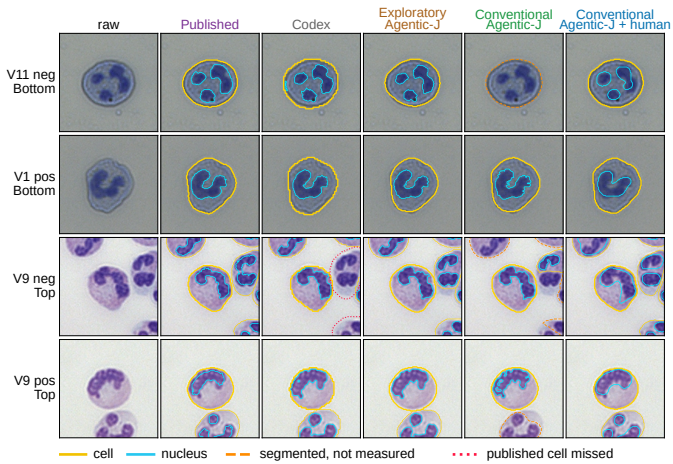
**a**

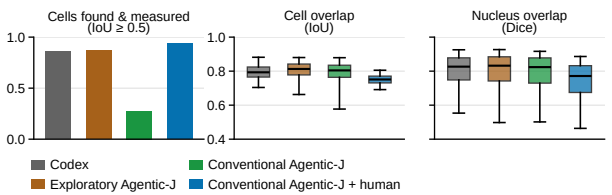
| | Analysis | CD177 neutrophil morphology: Time | CD177 neutrophil morphology: Scientific verdict | Leishmania infection: TUNEL and proliferation: Time T1 / T2 | Task 1 TUNEL calls (6 nuclei) | Task 2 published Fig. 4C (15 parasites) |
|---|---|---|---|---|---|---|
| Published | | weeks | no meaningful difference | not recorded | 2/6 nuclei TUNEL+ | 8 TUNEL+ vs 7 TUNEL−, p = 0.0012 |
| Codex CLI | | 13 min | ✓ 0/30 features significant | 18 min (both) | ✓ 6/6 calls, r = 0.98 | ✓ 10/15 found, hosts right; p = 0.017 |
| Exploratory Agentic-J | | 35 min | ✗ 5/25 significant (3/3 runs) | 22 / 69 min | ✓ 5/6 calls, r = 0.95, 1 split | ✗ 0/15 found (+22), no host |
| Conventional Agentic-J | | 67 min | ✓ 0/32 significant (3/3 runs) | 55 / 85 min | ✓ 5/6 calls, r = 0.99 | ✗ 3/15 found (+2), no TUNEL+ |
| Conventional Agentic-J + human | | 488 min | ✓ 0/11 features significant | 32 min / n.r. | ✓ 5/6 calls, r = 0.97 | ✓ 15/15; 14/15 within 0.1; p = 0.0012 |

Manual (analyst) · Human feedback (user) · Automatic (Agentic-J) · Automatic (other agent) · ✓ published conclusion reproduced · ✗ not reproduced

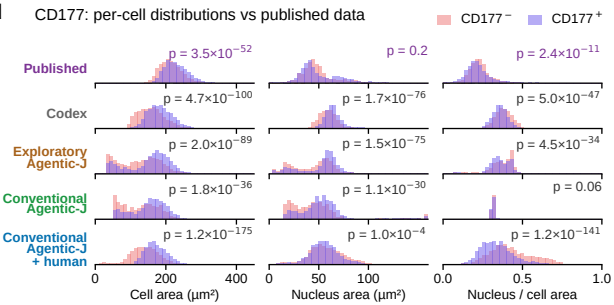
**b** CD177: segmentations of four example cells



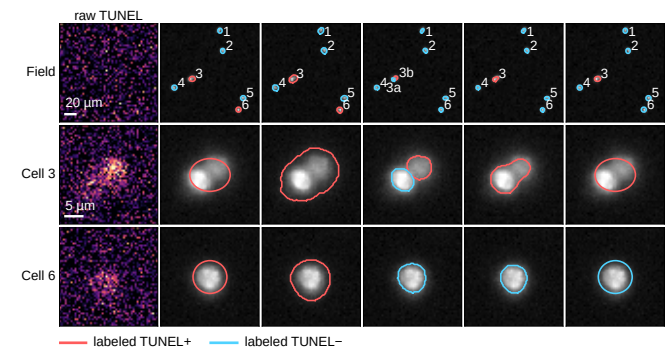

**c** CD177: agreement with the published masks (4389 cells)



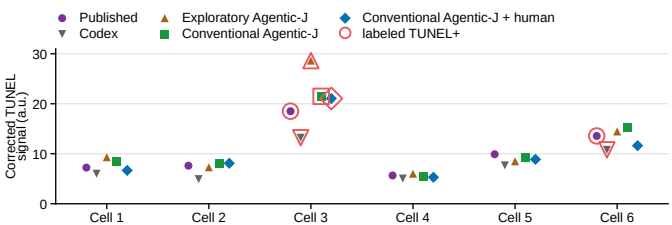

**d** CD177: per-cell distributions vs published data



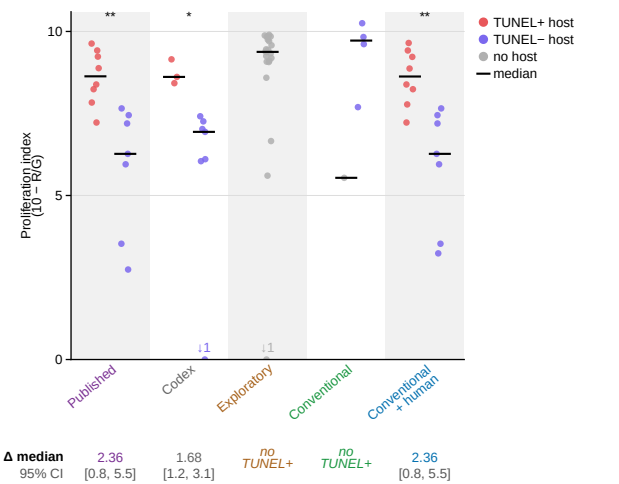

**e** Leishmania task 1: the same six nuclei, called by each pipeline




**f** Leishmania task 1: corrected TUNEL signal per nucleus




**g** Leishmania task 2: proliferation index (published Fig. 4C)




| Δ median (95% CI) | Published | Codex | Exploratory | Conventional | Conventional + human |
|---|---|---|---|---|---|
| | 2.36 [0.8, 5.5] | 1.68 [1.2, 3.1] | *no TUNEL+* | *no TUNEL+* | 2.36 [0.8, 5.5] |